

4.7.2 The motor effect

4.7.2.1 Electromagnetism

Content	Key opportunities for skills development
When a current flows through a conducting wire a magnetic field is produced around the wire. The strength of the magnetic field depends on the current through the wire and the distance from the wire. Shaping a wire to form a solenoid increases the strength of the magnetic field created by a current through the wire. The magnetic field inside a solenoid is strong and uniform. The magnetic field around a solenoid has a similar shape to that of a bar magnet. Adding an iron core increases the strength of the magnetic field of a solenoid. An electromagnet is a solenoid with an iron core. Students should be able to:	
• describe how the magnetic effect of a current can be demonstrated • draw the magnetic field pattern for a straight wire carrying a current and for a solenoid (showing the direction of the field) • explain how a solenoid arrangement can increase the magnetic effect of the current.	WS 2.2
(Physics only) Students should be able to interpret diagrams of electromagnetic devices in order to explain how they work.	(Physics only) WS 1.4

4.7.2.2 Fleming's left-hand rule (HT only)

Content	Key opportunities for skills development
When a conductor carrying a current is placed in a magnetic field the magnet producing the field and the conductor exert a force on each other. This is called the motor effect. Students should be able to show that Fleming's left-hand rule represents the relative orientation of the force, the current in the conductor and the magnetic field. Students should be able to recall the factors that affect the size of the force on the conductor.	

Content	Key opportunities for skills development
For a conductor at right angles to a magnetic field and carrying a current: <i>force = magnetic flux density × current × length</i> $[F = B I l]$ force, F, in newtons, N magnetic flux density, B, in tesla, T current, I, in amperes, A (amp is acceptable for ampere) length, l, in metres, m	MS 3b, c Students should be able to apply this equation which is given on the <i>physics equation sheet</i>.

4.7.2.3 Electric motors (HT only)

Content	Key opportunities for skills development
A coil of wire carrying a current in a magnetic field tends to rotate. This is the basis of an electric motor. Students should be able to explain how the force on a conductor in a magnetic field causes the rotation of the coil in an electric motor.	

4.7.2.4 Loudspeakers (physics only) (HT only)

Content	Key opportunities for skills development
Loudspeakers and headphones use the motor effect to convert variations in current in electrical circuits to the pressure variations in sound waves. Students should be able to explain how a moving-coil loudspeaker and headphones work.	